

1. Roadmap

📺 GenAI Roadmap for Beginners | End-to-End GenAI Course 2025 | CampusX

Introduction

GenAI (Generative AI) is a type of AI that learns patterns, mimics humans, and generates content. Content = text, image, video, audio, etc.

History:

- Research on AI has been conducted for the last 60-70 years.
- Technologies such as:
 - Symbolic AI: AI that works using rules and symbols (logic) instead of learning from data
 - Fuzzy Logics:
 - Evolutionary Algorithms:
 - NLP:
 - Computer Vision:
- Of these, ML has the largest impact.

Machine Learning (ML)

In ML, we provide a lot of data to a mathematical or statistical model algorithm. The algorithm then predicts new outputs.

Now, this simple idea was so powerful that it is also relevant today. There are ML jobs today, too.

For example:

- Stock price prediction is a type of regression problem.
- Cat or Dog Photo prediction is a type of classification problem
- Ranking products in recommendation systems.

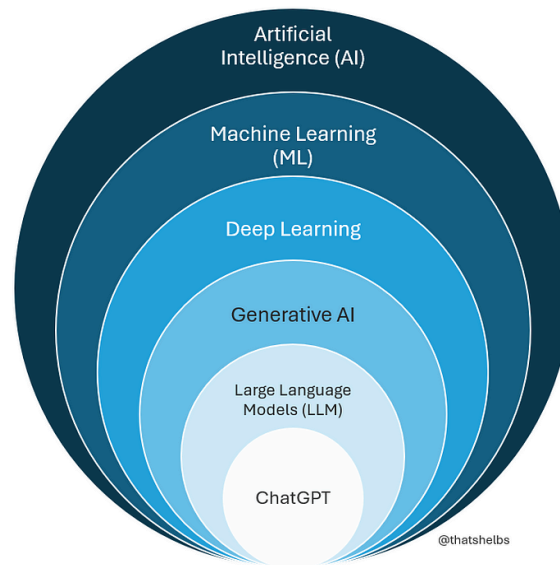
Meanings of words:

1. Regression = Output is numerical. age, price, temperature, etc.
2. Classification = Output is categorical. true/false, positive/negative, email is spam or not spam, etc.
3. Recommendation Systems = suggestion systems based on history

GenAI in AI Landscape:

ML is used to predict things only. GenAI is used for creating new things, too.

1. AI is the outermost circle. Everything comes under this umbrella. It has many things, like fuzzy logic, symbolic AI, etc.
2. ML is the most powerful technique. It has many things like statistical techniques, model-based techniques, etc.
3. Deep Learning uses Neural Networks and Transformers.
4. Neural Network + Transformers = GenAI.



GenAI Impact Areas:

- Customer Support: Using a Chatbot model
- Content Creation
- Education
- Software Development

Mental Model of GenAI (According to CampusX)

Foundation Models (Core Part)

- We can also say ML models.
- Foundation models are the very large AI models.
- They need very large data, very large hardware, so much money, and so many GPUs.
- These models are generalised, not specialised.
 - Specialised models mean the models are task-specific. For example, for stock price prediction, we make a special model that can do only that.
 - Generalised models mean the model can do multiple tasks because they get large data, the architecture is huge, and so many parameters are involved.
- An example of a foundation model is **LLMs**.
 - LLMs (Large Language Models) are the backbone of GenAI
 - LLMs can generate text, perform sentiment analysis, summarization, Q&A, predict the next word, etc.
 - LLMs have so many parameters, and they are trained on data from all over the internet.
 - LMMs are a type of LLMs that also generate images and videos.
- Thus, foundation models are the center of GenAI.

Builder's Perspective of Foundation Model

- Transformers Architecture
- Types of Transformers
 - Encoder Only
 - Decoder Only

- Encoder & Decoder Based (T5)
- Pretraining
 - Training Objectives
 - Tokenization Objectives
 - Training Strategies
 - Handling Challenges
- Optimization (try to reduce the size)
 - Training Optimization
 - Model Compression
 - Optimization Inference
 - Quantization
- Fine Tuning
 - Task Specific Tuning
 - Instruction Tuning
 - Continual Pretraining
 - RLHF
 - PEFT
- Evaluation
- Deployment

Users Perspective of Foundation Model

- Building Basic LLM Apps
 - Open Source vs Closed Source LLMs
 - Using LLMs APIs
 - LangChain
 - HuggingFace
 - Ollama

NOTE: Some of the above are API-based approaches, while some can run on our local machine.

- Improve outputs
 - Prompt Engineering
 - Art and Science of writing prompts
 - Prompts = our inputs for the foundation models
 - RAG
 - Show our private data, and Q&A

- Fine Tuning
 - Compared to the builder's perspective, this is less technical here.
- Build Agents
 - Provide tools to AI
 - Not only produce output, but also perform actual tasks.
 - Example: Not only find the best tourist locations, but also automatically book flight tickets.
- LLMOps
 - The complete process from the user's perspective in Engineering
- Miscellaneous (General)